

ForeDance: Structured Streaming Dance Generation with Anticipatory Music Conditioning

Abstract—Streaming dance must prepare for upcoming music from audio heard so far while continuing movements that have already been committed and can no longer be revised. We present ForeDance, a framework that generates robot-native dance from music as it arrives. A frozen music foundation model forecasts future musical context using heard audio alone. A kinematics-guided structure–detail representation gives whole-body organization and local articulation complementary roles, while Commit Forcing trains continuation from jointly generated motion and its resulting reference state. Together, these components couple musical anticipation with successive motion decisions. On 60-second FineDance generations, a frozen music–motion evaluator favors the intended soundtrack over alternative tracks in approximately 75% of pairwise comparisons. Separate matched training comparisons identify a tradeoff between motion distribution fidelity and smoothness. These findings support soundtrack correspondence in continuously generated references within a framework that connects musical foresight to committed movement.

I. INTRODUCTION

Dance preparation begins before a visible accent: weight shifts precede a step, and the body transitions toward a pose before reaching it. A system dancing to music as it arrives must choose these movements before hearing all the sound they will accompany. Waiting for an event leaves less time to prepare; reading the future recording changes the task into offline generation. Streaming further requires each new movement to continue from emitted motion that can no longer be revised. Musical anticipation and motion continuation are therefore coupled requirements of streaming dance generation.

Music-conditioned choreography has advanced through expressive motion models and large paired datasets [1]–[4]. Streaming systems such as DiscoForcing [5] encode incoming audio causally, supporting motion generation as the soundtrack unfolds. Our question is how to supply prospective musical context when future audio is unavailable. We investigate this question in robot-native dance: the model produces reference motion in the humanoid’s own configuration space, so musical choices must be expressed through its body and continued across successive planning decisions.

We present ForeDance, a framework that conditions streaming dance on predicted future music states. A frozen Magenta RealTime 2 (MRT-2) music foundation model [6] rolls forward from the audio received so far. This transfers a general musical continuation prior into dance generation: paired dance data teaches the motion model to use the prior, while musical prediction is learned through large-scale music pretraining. To our knowledge, ForeDance is the first streaming dance generation framework to explicitly condition

motion on future music states forecast by a pretrained music foundation model using past and present audio alone. The predicted states guide both whole-body organization and complementary articulation.

Realizing this design requires a representation that organizes movement and a training procedure that accounts for earlier generated choices. Our kinematics-guided structure–detail representation couples discrete structure and continuous detail with complementary reconstruction objectives. Structure-only reconstruction preserves designated body organization and endpoint placement; their combination reconstructs full motion through a shared decoder. Building on hybrid motion representations [7], this supervision specifies what the two components should preserve. Commit Forcing then trains continuation after jointly sampling and committing both components. Their decoded movement determines the next starting state, pairing generated history with the reference state it actually induces.

At each decision, ForeDance plans a short segment, commits its first segment, and updates the context for the next plan (Fig. 1). A reference interface delivers the resulting stream to a fixed SONIC whole-body controller [8]. The generator continues from committed references; measured robot feedback is handled by the controller. This keeps the motion-generation mechanism explicit while providing a path to motion tracking.

Our contributions are:

- **Anticipatory music conditioning for streaming dance.** Predicted future states from a frozen music foundation model provide musical lookahead using arrived audio alone.
- **Structure–detail representation and paired training.** Complementary reconstruction objectives establish structural and detail roles for coupled music-conditioned generation through a shared decoder.
- **Commit Forcing for streaming continuation.** Jointly generated structure and detail define both the committed history and its resulting reference state for continuation training.

Evaluations of 60s FineDance generations find that a frozen music–motion evaluator favors the intended soundtrack in approximately 75% of pairwise comparisons. A separate, complete 18-track study characterizes the effects of history regularization on motion fidelity, diversity, and smoothness. Together with rendered references, these studies examine both musical correspondence and the resulting motion. Section V distinguishes these findings from the

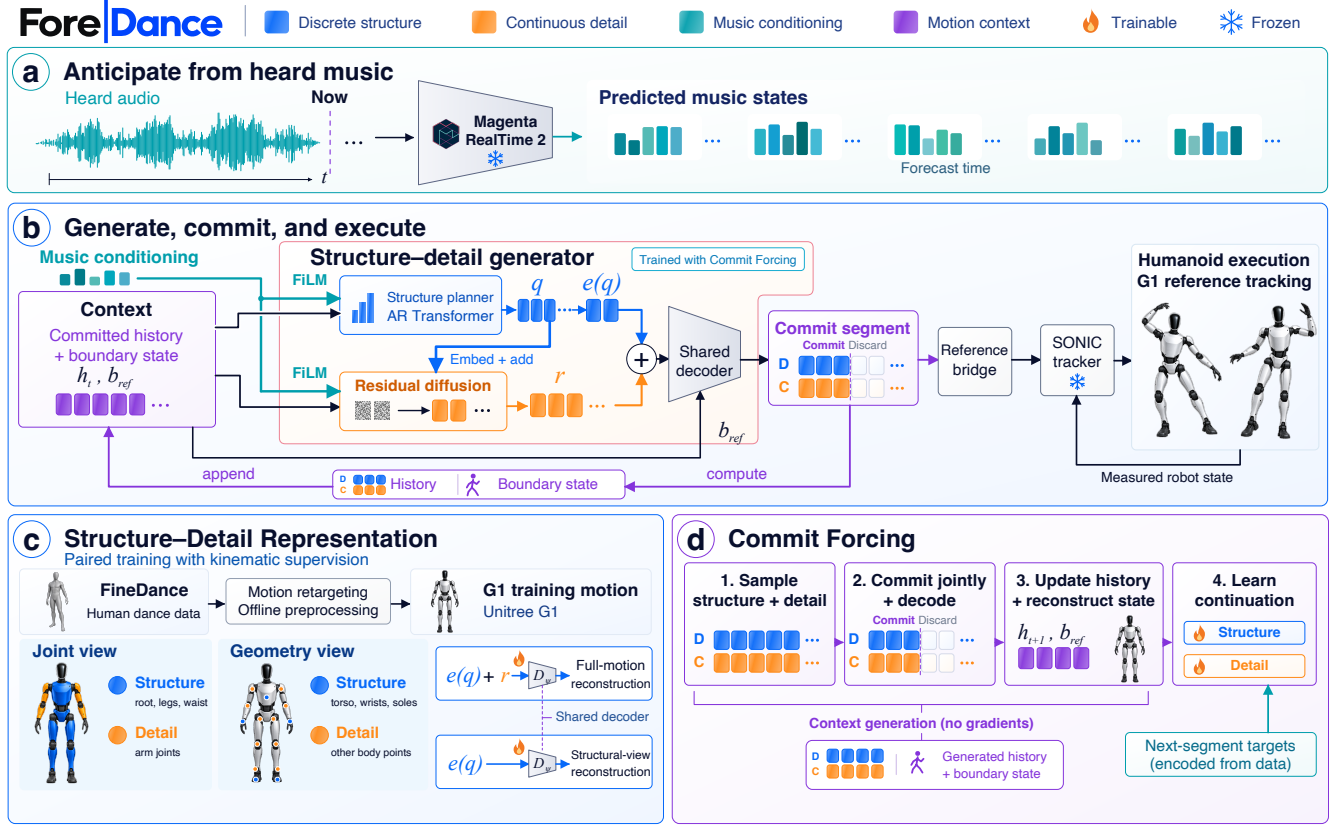


Fig. 1. **ForeDance overview.** **a:** Heard audio supplies a causal forecast of upcoming music states. **b:** The music condition enters both generation branches through FiLM; sampled structure also conditions detail. A shared decoder maps their sum to reference motion. Joint commitment updates code history and the reference-derived boundary state, while measured robot feedback stays within the fixed tracking loop. **c:** Paired reconstruction gives structure and detail complementary kinematic roles. **d:** Commit Forcing trains continuation from jointly generated history and its resulting state. Robot poses and time ranges are schematic, not experimental or hardware execution evidence.

comparisons needed to attribute forecast and component benefits.

II. RELATED WORK

A. Observed and Predicted Music Conditions

Music-to-dance models combine acoustic descriptors [9] or learned features [10], [11] with choreography models. FACT and Bailando model coordinated sequences and choreographic memory [1], [12]; EDGE and LODGE develop diffusion-based and long-form generation [2], [13]. FineDance, MEGADance, and SoulNet address fine-grained motion, genre, and music–motion alignment [3], [4], [14].

DiscoForcing [5] uses a task-specific causal VQ-PAE encoder to describe heard audio through rhythmic tokens and periodic alignment. We instead condition on explicit future states from a generative music prior. MRT-2 is pretrained on approximately 71,000 hours of music [6], providing continuation knowledge beyond our paired dance training. Foundation-model features already support dance in EDGE through Jukebox [2], [15] and in DGFM [16]; earlier robot systems also predict beat timing [17]. Our distinction is the use of generative music-state forecasts under a causal audio constraint. Live Music Models [18] and MRT-2 supply the

music-generation foundation; we transfer its predictions to structured streaming motion.

B. Structure–Detail Representation and Training

Discrete tokens support motion generation in T2M-GPT and MoMask [19]–[21]; MLD uses continuous latent diffusion [22], and DisCoRD uses flow-based decoding from discrete tokens [23]. Most closely, DC-Motion [7] combines discrete structure with continuous residuals. We build on this hybrid principle, using designated robot-kinematic supervision to give the components complementary roles through a shared, boundary-conditioned decoder. Neural Categorical Priors and GPC learn vocabularies for physical control [24], [25]; our latents instead decode into explicit motion references for a separate tracker.

C. Continuation from Generated Context

Scheduled Sampling and Professor Forcing address training–sampling mismatch [26], [27]. Diffusion Forcing varies per-token noise [28]; MotionStream and MotionStreamer develop causal motion generation [29], [30]. MotionStreamer’s Two-Forward strategy uses first-pass predictions in second-pass history. Commit Forcing constructs the joint structure–detail history–state transition: sampled components enter history together, and their decoded committed

motion determines the next reference state. Its distinction is this coupled update. Diffusion Policy and Real-Time Chunking address committed actions during plan updates [31], [32]; CoF constructs the context under which continuation is learned.

a) Robot execution: Reusable trackers support expressive motion [8], [33]–[35], while HARMON and RoboPerfom introduce language and audio conditioning [36], [37]. PhysDiff, terrain-aware reference generation, and PARC incorporate physical correction or adaptation [38]–[40]. Our fixed-controller interface keeps reference generation separate from measured-state feedback, allowing their outputs to be examined independently.

III. METHOD

ForeDance links musical forecasting, structured motion generation, and continuation from committed motion. Figure 1 shows the common forecast condition and the motion context shared by these operations.

a) Streaming task: At a planning decision, $a_{\leq \tau}$ is the arrived audio, h is committed motion-code history, and b is the pose and velocity state derived from the last committed reference frames. A frozen music model produces future states $m^{\text{pred}} = F(a_{\leq \tau})$. The generator samples

$$(q_{1:H}, r_{1:H}) \sim p_{\theta}(q, r \mid h, b, m^{\text{pred}}), \quad (1)$$

where q and r are discrete structure and continuous detail. A shared decoder maps them to native G1 motion. We commit the first $C < H$ latent steps, discard the remaining plan, and update history and the reference state. Thus later decisions may revise uncommitted plans while preserving the emitted motion.

A. Anticipatory Music Conditioning

The next movement will accompany sound that is only partly observed. We provide prospective context by tokenizing arrived audio with SpectroStream and rolling frozen MRT-2 Small forward [6]. Its general music pretraining supplies the continuation prior; paired dance training learns how to translate the predicted states into movement. The forecast covers approximately 0.56 s, alongside a motion plan of approximately 0.53 s. No future waveform samples enter the condition.

Each forecast state carries its intended time. Learned projections and time embeddings align these states to motion positions, and frame-wise feature modulation [41] conditions both branches:

$$\text{FiLM}(v_i, c_i) = \gamma(c_i) \odot v_i + \beta(c_i). \quad (2)$$

Here v_i is a generator activation, c_i its aligned music condition, and γ, β predict a scale and shift. Conditioning both branches lets the same prospective context guide body organization and local articulation; structure also conditions the detail generator. The music model stays frozen while these motion-side mappings are learned. Expired forecast states are masked by their timestamps. Appendix I specifies timing and conditioning details.

B. Structure–Detail Representation and Paired Training

Musically directed motion needs coordinated body organization together with local articulation. We learn these roles through complementary supervision, using a discrete–continuous latent

$$z_i = e(q_i) + r_i, \quad (3)$$

where $e(q_i)$ is a codebook embedding and r_i a continuous residual. A shared decoder $D_{\psi}(z, b)$ maps their sum and starting state to robot-native motion. Quantization supplies a structural hypothesis; the residual adjusts movement within it. The training objectives determine which properties this division should preserve (Fig. 2).

a) Learning complementary roles: We pair full-latent reconstruction $D_{\psi}(e + r, b)$ with structure-only reconstruction $D_{\psi}(e, b)$. The full path reconstructs all motion views, whereas the structure-only path is supervised on designated structural views. Both use the same decoder and starting state. This gives discrete structure an explicit kinematic responsibility while allowing the combined latent to retain complete movement.

The supervision views cover native coordinates and velocities, plus forward-kinematics positions and velocities. In joint space, structure selects root, legs, and waist; detail contains the arm coordinates. In geometry, structure selects the torso, both wrists, and both sole points; detail contains the remaining body points. Thus wrist placement can be structural while arm-joint articulation belongs to the complementary native view. These are supervision views of shared decoding, not separate anatomical decoders. An auxiliary structural loss under residual corruption reinforces the same organization when detail varies. Exact partitions, reconstruction gradients, and loss weights are given in Appendix I.

b) Coupled motion generation: A causal Transformer samples structural tokens. A residual diffusion generator [42] conditions on that sampled structure, the common history and starting state, and the forecast music. Their sum decodes into 38-dimensional G1 motion, including root movement and 29 joint angles. Both branches commit the same time segment, preserving their pairing across plans. The next segment starts from a reference-derived pose and velocity state; the coordinate convention and complete state definition are in Appendix I.

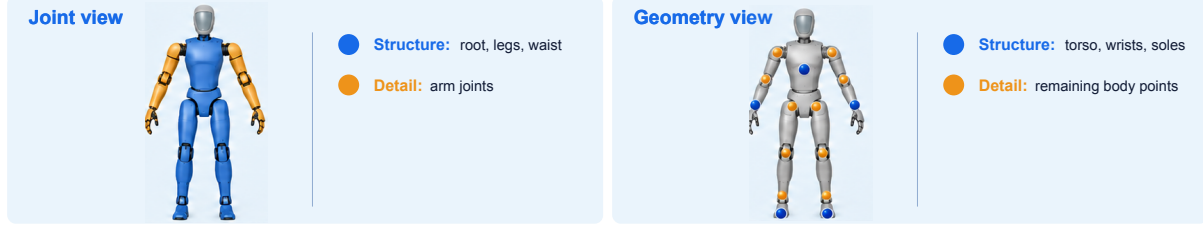
C. Commit Forcing

Every committed segment changes both latent history and the reference state from which the next segment begins. Teacher Forcing (TF) supplies these inputs from recorded motion; streaming inference obtains them from generated motion. Commit Forcing constructs this generated transition before supervising continuation (Fig. 3). A stop-gradient first pass samples structure $\tilde{q}$ and detail $\tilde{r}$ with the inference samplers. Their jointly committed segment is decoded, and an update operator U constructs

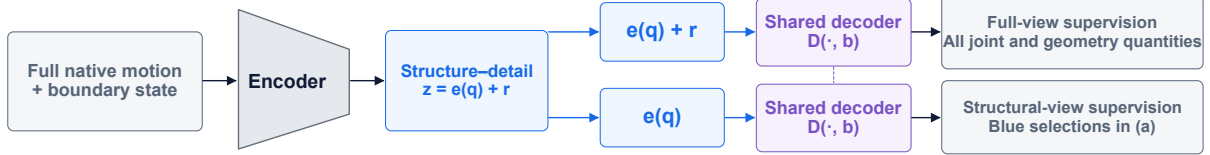
$$(\tilde{h}^+, \tilde{b}^+) = U(h, b, \tilde{q}_{1:C}, \tilde{r}_{1:C}). \quad (4)$$

Structure–Detail Representation and Paired Training

(a) Body-part supervision views



(b) Two principal reconstruction paths



Supervision views guide latent roles; both paths share one decoder.
Full reconstruction also uses boundary-transition supervision.

Fig. 2. **The representation and its training are one design.** Body-part supervision has separate joint and geometry partitions: root/legs/waist versus arms in joint space, and torso/wrists/soles versus other body points in geometry. The two principal reconstruction paths supervise the full latent on all views and discrete structure on the structural view. The shared decoder and boundary state are used throughout.

The next training pass receives the appended codes together with the state induced by their movement. This keeps history and state consistent with the same generated commitment rather than combining unrelated transitions.

a) Learning the next plan: The next recorded latent remains the training target. For target $z^* = e(q^*) + r^*$ and sampled next structure $\hat{q}$, we rebase the residual:

$$r_{\text{rebased}}^* = z^* - e(\hat{q}). \quad (5)$$

Consequently, $e(\hat{q}) + r_{\text{rebased}}^* = z^*$: rebasing preserves the target latent. Decoded movement also depends on the starting state and need not be identical under recorded and generated states. Each example independently constructs its first-pass transition; the procedure trains continuation after a generated commit without requiring a fully recursive training rollout. The distinguishing operation is the joint history–state update, which exposes the learner to the coupled consequences of its structure and detail predictions.

b) Supporting history regularization: Full-history corruption (FHC) perturbs discrete embeddings and residuals across the motion history, after CoF context construction. It leaves the reference state and music condition unchanged and is disabled at inference. Appendix I gives its settings.

D. Robot Reference Integration and Execution

Figure 4 separates reference delivery from controller feedback. The committed native motion provides the reference interface to a separate, pretrained SONIC whole-body controller [8]. A reference bridge reconstructs root motion in the fixed sequence coordinate system, maps joint references into the controller convention, and resamples the 30 Hz motion onto a continuous 50 Hz grid. Appendix I defines the coordinates and state reconstruction. Only committed motion

is delivered; the uncommitted suffix is discarded before the next plan and is not sent to the controller.

SONIC remains fixed and combines the consumed reference with measured robot state to produce control actions. The generator separately updates its latent history and boundary state from committed reference motion. Appendix IV specifies the delivery and measurement protocol.

IV. EXPERIMENTS

We separate evaluation of the music–motion evaluator on reference motion from its use to assess generated dance. Beat alignment and motion-quality measures complement learned correspondence. Supporting history-training comparisons examine motion quality; the core ablations still awaiting results are marked explicitly in Section IV-D.

A. Setup and Metrics

Evaluator validation uses reference motion from 30 content groups, comprising 3,661 windows and 15 disjoint song pairs per discrimination task. Generator correspondence uses 99 generated 60 s FineDance trajectories: eleven tracks, three independently trained generators, and three sampling seeds per generator. The separate beat-alignment and motion-quality study uses all 18 archived tracks, a common 23 s segment, and the same three-by-three training/sampling repeat structure, giving 162 outputs per configuration. All motion is represented in native G1 space. Appendices I and II specify the training and cohort protocols.

a) Learned music–motion correspondence: Motivated by music–motion retrieval [4], we use a frozen G1-native evaluator trained independently of the generator. Its score $s_w(x, a)$ averages cosine similarities from three evaluator instances for five-second music–motion windows. For intended

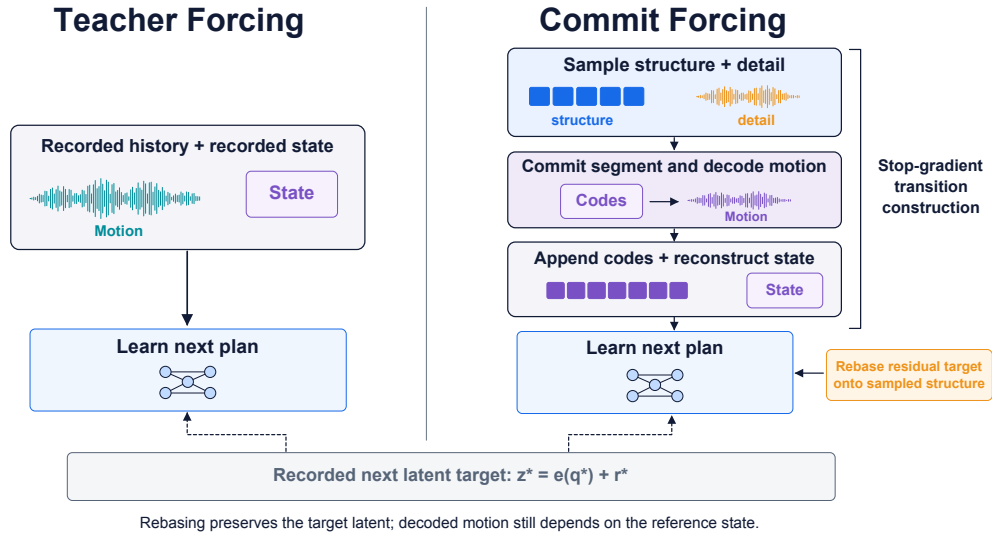


Fig. 3. **Learning the next plan: TF versus CoF.** TF receives a recorded history–state pair. CoF samples and commits structure and detail together, derives the corresponding next reference state, and supervises continuation. Both use recorded targets. Residual rebasing retains the target latent under sampled structure; decoding additionally depends on the supplied boundary state.

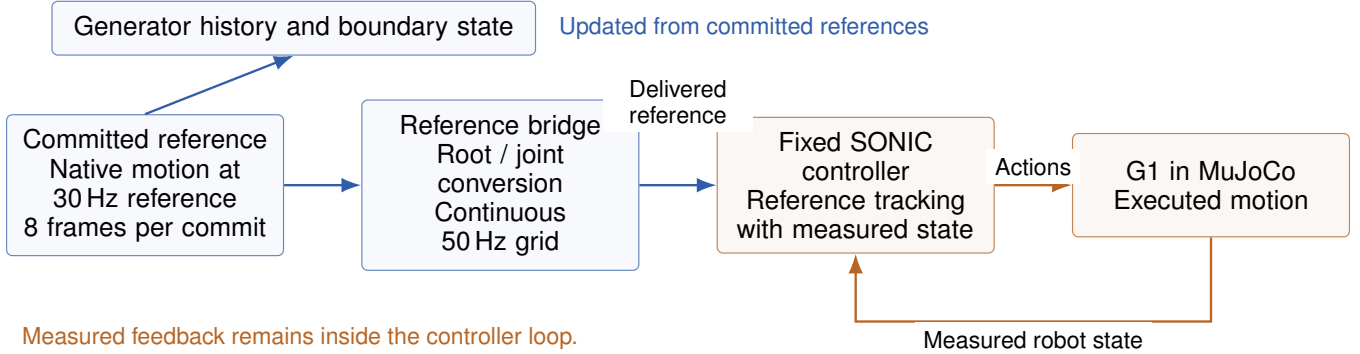


Fig. 4. **From committed references to motion tracking.** The bridge converts the reference stream onto the controller grid. Committed references update the generator context, whereas measured robot state closes the SONIC tracking loop.

music a_i and an alternative track a_j , the correspondence margin is

$$\delta_{ijw} = s_w(x_i, a_i) - s_w(x_i, a_j). \quad (6)$$

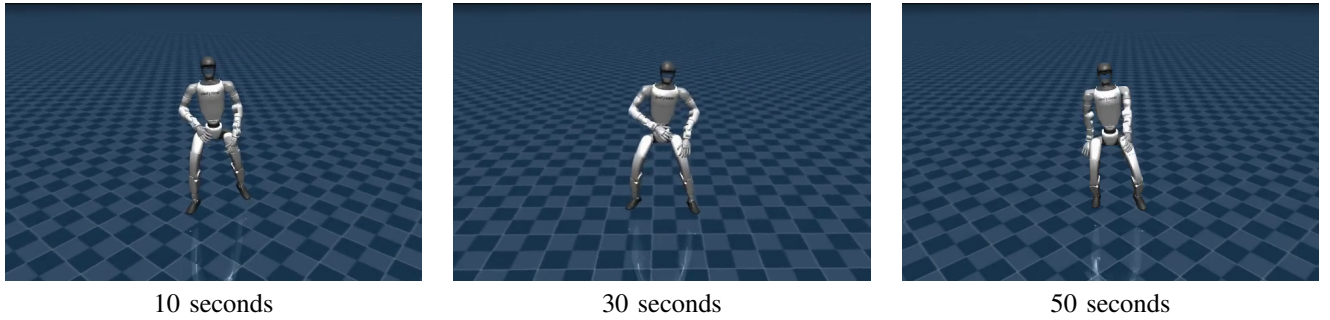
Pairwise accuracy counts positive margins, with half credit for ties; random ordering gives 50%. The evaluator-validation tasks use ordinary or musically similar alternative tracks. Generator scoring compares each fixed motion against its intended music and all ten alternatives in its fixed gallery, preserving native audio offsets. Windows and sampling repeats are aggregated within training-seed/query-track cells. Appendix II-A defines the frozen evaluator.

b) Beat alignment and motion quality: Beat Alignment Score (BAS) measures the proximity of motion beats to music beats. We use local minima of smoothed mean forward-kinematics keypoint speed and a motion-to-music Gaussian matching score with a 0.10 s scale; Appendix II-A.0.e gives the exact definition. BAS measures rhythmic timing, while learned correspondence assesses soundtrack preference. Ki-

TABLE I
EVALUATOR VALIDATION ON REFERENCE MOTION. THE FROZEN EVALUATOR RANKS INTENDED MUSIC AGAINST ORDINARY OR MUSICALLY SIMILAR ALTERNATIVES. PAIRWISE ACCURACY IS IN PERCENT; RANDOM ORDERING IS 50%.

Discrimination task	Accuracy (%)
Ordinary alternative music	68.41
Similar alternative music	64.08

netic and geometric Fréchet distances (FID_k , FID_g) compare generated and reference distributions of motion dynamics and body configurations. Div_k and Div_g measure spread in the same feature spaces. PFC is a contact-related physical-consistency proxy, complemented by root/contact and jerk diagnostics. The 23 s and 60 s cohorts are analyzed separately.



ForeDance: selected 60-second FineDance generation.

Fig. 5. A 60-second ForeDance generation. Original frames at 10, 30, and 50 s from a selected FineDance sequence. The images show rendered reference motion.

TABLE II

MUSIC ALIGNMENT OF FOREDANCE GENERATIONS. LEARNED CORRESPONDENCE IS PAIRWISE SOUNDTRACK ACCURACY; BAS MEASURES MOTION-TO-MUSIC BEAT ALIGNMENT. HIGHER IS FAVORABLE FOR BOTH METRICS; THEIR EVALUATION COHORTS AND SCORE SCALES DIFFER.

Measure	Evaluation cohort	Score
Learned correspondence	11 tracks, 60 s	75.01%
Beat alignment (BAS)	18 tracks, 23 s	0.4419

B. Evaluator Validation

Table I evaluates the score itself on reference motion. Pairwise accuracy is 68.41% against ordinary alternative music and 64.08% against musically similar alternatives. Both exceed random ordering in point estimate, with the similar-music task providing a more demanding correspondence check. These results characterize the evaluator’s ability to distinguish soundtracks, rather than the quality of ForeDance generations.

C. Generator Evaluation

Table II reports learned soundtrack correspondence and beat alignment for generated motion. The intended soundtrack wins 75.01% of pairwise comparisons, with a mean correct-minus-alternative score margin of 0.1783. Every generator training seed has a positive mean margin. The separate 18-track study yields BAS 0.4419. These complementary measures characterize soundtrack preference and rhythmic alignment; their different cohorts preclude treating the values as a single pooled evaluation. Appendix III retains the detailed score statistics and descriptive intervals.

a) History-training comparison: Table III compares clean history, history dropout, and full-history corruption (FHC), holding the remaining settings fixed. Relative to clean history, FHC reduces kinetic/geometric FID by 21.9%/15.0%, with both directions favorable in all three training seeds, and increases kinetic diversity. BAS rises from 0.4352 to 0.4419; this is a descriptive change, not a significance claim. PFC and jerk favor clean history, so the distribution gains coexist with a smoothness tradeoff. This comparison examines history

regularization, rather than isolating the benefit of musical anticipation or Commit Forcing.

b) Qualitative generation: Figure 5 shows reference poses at 10, 30, and 50 s from one selected 60 s rollout. These original frames illustrate body configuration across the sequence; they do not measure temporal smoothness.

D. Pending Ablation Studies

TODO for this working draft: the following controlled studies await results and are not counted as completed evidence.

a) TODO: musical anticipation: Compare forecast-conditioned and heard-music-conditioned generators with matched training, tracks, and sampling. Report learned correspondence, BAS, and motion quality to isolate the value of predictive musical context.

b) TODO: paired representation training: Complete current-representation reconstruction and intervention results to test whether structure and detail serve their intended complementary roles. Include non-collapse checks and matched supervision comparisons.

c) TODO: single-representation controls: Compare independently trained discrete-only, continuous-only, and combined structure–detail generators under matched settings. Removing a decoder branch does not substitute for a separately trained baseline.

d) TODO: committed continuation: Compare Commit Forcing with Teacher Forcing on matched 60 s free-running sequences. Report aggregate quality and each of the three non-overlapping 20 s windows, including commitment-boundary behavior and matched qualitative sequences.

E. Supporting Execution Diagnostics

The interface in Section III-D separates the generated, resampled, controller-consumed and measured trajectories. The supporting study replays four saved trajectories from one FineDance track through fixed SONIC in MuJoCo. These trajectories use an earlier 34-dimensional representation, separate from the current 38-dimensional generation cohort; audio inference is not run during replay.

After excluding the first two seconds and aligning joint trajectories, median amplitude retention ranges from 0.858

TABLE III

MOTION QUALITY AND BEAT ALIGNMENT UNDER MATCHED HISTORY TRAINING. ALL THREE CONFIGURATIONS SHARE INITIALIZATION, ARCHITECTURE, MUSIC CONDITIONING, THE COMPLETE 18-TRACK COHORT, COMMON 23-SECOND SEGMENTS, THREE TRAINING SEEDS, AND THREE SAMPLING SEEDS. LOWER IS FAVORABLE FOR FID AND PFC; HIGHER BAS INDICATES CLOSER BEAT ALIGNMENT. DIVERSITY DESCRIBES MOTION SPREAD AND IS INTERPRETED TOGETHER WITH FIDELITY.

History training	FID _k	FID _g	Div _k	Div _g	BAS	PFC
Clean history	208.599	0.482	7.495	0.493	0.4352	0.143
History dropout	164.761	0.469	7.801	0.473	0.4338	0.182
History corruption	162.831	0.410	8.626	0.508	0.4419	0.167

to 0.901, and median squared-velocity retention from 0.411 to 0.496. Position tracking therefore coexists with dynamic attenuation. These diagnostics characterize the reference-to-execution path; they do not establish current-model hardware performance or comparative musical benefit. Appendix IV retains the complete measurements and protocol.

V. DISCUSSION

a) Supported findings and attribution: The generator studies characterize soundtrack correspondence, beat alignment, and a distribution–smoothness tradeoff in history regularization. The evaluator’s reference-motion validation is a separate measurement. Attributing gains to anticipation, representation design, or committed continuation requires the controlled studies listed in Section IV-D. Broader music pretraining supplies a transferable prediction prior, while improved out-of-distribution dance generalization remains a hypothesis. The historical replay diagnostics characterize tracking separately from these current reference-generation results.

b) Evaluation scope and selection: The evaluator configuration was selected after inspection of its 30-content-group calibration cohort. For compact reporting, generator music scores use the higher complete initialization result for each metric after evaluation; all tracks and sampling repeats in the respective cohort are retained. The history-training comparison uses one shared initialization. Calibration scores and retained intervals are descriptive, not selection-independent evidence of superiority. The score has not been validated against human preference or executed trajectories. The motion-quality cohort is separate, and the illustrated sequence is a selected qualitative case. Its complete source and selection record are retained with the evidence.

VI. CONCLUSION

ForeDance brings anticipatory music conditioning to structured streaming dance: a frozen music foundation model predicts prospective context, kinematics-guided structure–detail training organizes motion components, and Commit Forcing learns continuation after their joint commitment. Completed studies find consistent soundtrack correspondence and characterize motion quality under matched history-training choices. Together, the method and reference interface provide an explicit framework for studying how musical foresight translates into continuously generated robot motion.

APPENDIX I

REPRESENTATION AND TRAINING DETAILS

a) Native representation and boundary state: We represent motion directly in the robot’s configuration space, including full pelvis orientation. A fixed yaw anchor R_A is removed once for a motion sequence, and every frame stores absolute orientation in that anchor:

$$R_t^A = R_A^\top R_t^{\text{world}}, \quad (7)$$

$$x_t = [\Delta p_{xy,t}^{\text{local}}, h_t, \rho(R_t^A), \theta_t]. \quad (8)$$

Here $\Delta p_{xy,t}^{\text{local}}$ denotes planar displacement, h_t is pelvis height, ρ is the six-dimensional rotation representation [43], and θ_t contains 29 joint angles, giving 38 motion dimensions. This retains roll and pitch without multiplying a sequence of predicted rotation increments. Planar displacements are integrated using the appropriate preceding heading. The anchor remains fixed over the generated sequence; it is not reset at every commit.

The corresponding 71-dimensional boundary state is

$$b_t = [h_t, v_{xy,t}^{\text{local}}, v_{z,t}, \rho(R_t^A), \omega_t, \theta_t, \dot{\theta}_t]. \quad (9)$$

The state contains pelvis height, local planar and vertical velocities, orientation, angular velocity ω_t , joint angles, and joint velocities. Conditioning on these quantities makes the starting pose and motion explicit at each segment boundary. They are reconstructed from the committed reference.

b) Streaming configuration: The motion rate is 30 Hz, with two frames per latent step. We use 64 history steps (128 frames, approximately 4.27 s), an eight-step plan (16 frames, approximately 0.53 s), and a four-step commit (eight frames, approximately 0.27 s). The codec latent has 16 dimensions and its codebook has 512 entries. The music condition uses 14 native forecast states at 25 Hz, spanning approximately 0.56 s. Each state retains its physical timestamp.

c) Reconstruction gradients and auxiliary objectives: Let sg denote stop-gradient. For encoder output z , embedding e , and $r = z - sg(e)$, we form full and structure-only reconstructions, together with an auxiliary residual-corrupted reconstruction:

$$x_f = D_\psi(sg(e) + r, b), \quad (10)$$

$$x_s = D_\psi(z + sg(e - z), b), \quad (11)$$

$$x_c = D_\psi(sg(e) + \mathcal{C}(r), b). \quad (12)$$

The full branch is supervised on Φ_f ; the other two are supervised on Φ_s . The corruption $\mathcal{C}$ either zeros detail or adds small channel-scaled noise, encouraging the structural reconstruction to persist under changes in the residual. The combined objective is

$$\mathcal{L}_{\text{codec}} = \mathcal{L}_f + \mathcal{L}_s + 0.25\mathcal{L}_{\text{robust}} + \mathcal{L}_{\text{VQ}} + \mathcal{L}_{\text{boundary}}. \quad (13)$$

The final term supervises boundary-transition derivatives and pelvis orientation at the beginning of the reconstructed segment. Appendix I provides the training details.

d) Conditioning and sampling: For forecast state u_j with lead ℓ_j relative to the motion decision, we form $c_j = \text{WLN}(u_j) + \phi(\ell_j)$, where W is a learned projection, LN denotes layer normalization, and ϕ represents physical time. Timestamp-based masking excludes expired forecast states from their aligned motion positions. A null condition bypasses music modulation. The structural planner uses a classification objective and the detail model a denoising objective. Residual generation uses ten DDIM steps [44].

e) Kinematic reconstruction losses: The full view contains four blocks: native motion, native velocity, forward-kinematics positions, and geometry velocity. Each block is standardized with training-split statistics. For a view Φ with blocks $\Phi^{(k)}$, define the reconstruction error

$$\mathcal{E}_{\Phi}(\hat{x}, x) = \frac{1}{4} \sum_{k=1}^4 \text{mean} \left[\left(\Phi^{(k)}(\hat{x}) - \Phi^{(k)}(x) \right)^2 \right]. \quad (14)$$

This gives $\mathcal{L}_f = \mathcal{E}_{\Phi_f}(x_f, x)$, $\mathcal{L}_s = \mathcal{E}_{\Phi_s}(x_s, x)$, and $\mathcal{L}_{\text{robust}} = \mathcal{E}_{\Phi_s}(x_c, x)$. Equal block weights prevent the number of geometric coordinates from determining their relative contribution. The VQ term is the codebook loss plus 0.25 times the commitment loss.

For the auxiliary robustness objective, the residual is zeroed with probability 0.5. Otherwise, Gaussian noise is added with scale sampled uniformly from 0.01–0.05 times the residual channel standard deviation. The resulting reconstruction is supervised on the structural view. This codec objective is separate from FHC, which regularizes the generator’s motion history.

The boundary loss supervises the first three decoded frames relative to the supplied boundary. It combines normalized smooth- L_1 errors for velocity, acceleration, and jerk with weights 1, 0.5, and 0.25, respectively, plus pelvis-rotation geodesic error with weight 1. These quantities explicitly penalize a reconstructed segment whose beginning is inconsistent with the target transition.

f) Body-part supervision views: In the native block, structure selects the first 24 of 38 coordinates: nine root coordinates, twelve leg joints, and three waist joints. The remaining 14 coordinates are the bilateral shoulder (three per arm), elbow (one), and wrist (three) joints. In the 35-dimensional velocity block, structure selects the first 21 coordinates: root linear/angular velocity and the same 15 leg/waist joint velocities. The other 14 are arm-joint velocities.

The geometry block contains 31 points: 29 non-root body origins and two lowest-foot geometry points. Structure selects the torso, the two wrist-yaw body origins, and the two lowest-foot points. Detail selects the remaining 26 points. Geometry velocity uses the same partition. These are supervision and analysis views, not independent anatomical decoders. In particular, the structural geometry view can constrain wrist placement while the complementary native view describes arm articulation.

g) Generator optimization and initialization: Generator training uses 6,250 updates with batch size 48, corresponding to 300,000 training windows. Mixed-start training draws 20% song-origin starts and 80% continuation windows; continuation-start training uses ordinary windows throughout. Residual sampling uses guidance fixed at 1.0; the music model remains frozen during generator training.

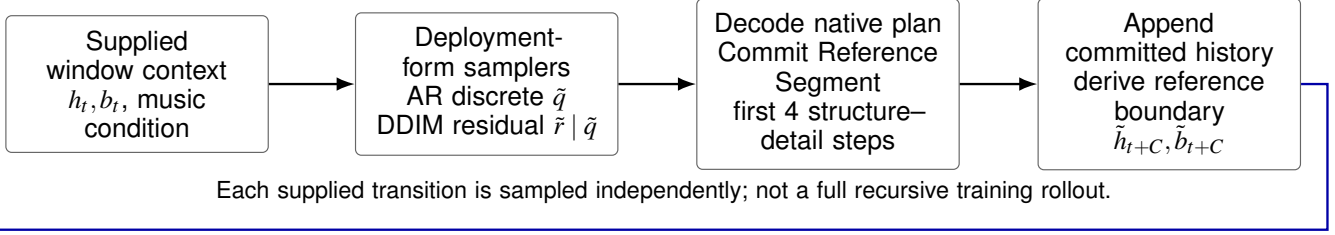
h) History regularization: FHC applies to the complete 64-step history with probability 0.5 per example. The discrete embeddings and continuous residual history share a severity timestep sampled from 850–999, with independent Gaussian noise. Their masks and timestamps are preserved, as are the boundary state and forecast conditions. Corruption is applied in the supervised pass after CoF context construction and is disabled at inference.

i) Constructing CoF transitions: Figure 6 details the construction. Each training example independently samples a first-pass transition from its supplied context. The committed structure and detail codes enter the history together, and their decoded motion determines the next boundary state. The next recorded structural and latent targets remain fixed. Residual rebasing expresses that latent target relative to the sampled structure; it does not re-encode the target under the generated boundary. This construction trains local continuation after a generated commit without requiring a fully recursive long training rollout.

APPENDIX II EVALUATION PROTOCOL

The reported correspondence result uses 99 trajectories: three generator training seeds, three sampling seeds, and eleven tracks. Each trajectory contains 1,800 frames at 30 Hz and supplies 56 overlapping five-second windows with one-second stride. For every motion, the intended audio is compared with all ten wrong-track donors at matched relative windows, preserving each source’s native audio offset. Means are formed over windows and donors, then sampling repeats within training-seed/query-track cells, and finally the three training seeds and 11 query tracks. Each bootstrap draw resamples three training seeds and eleven query tracks with replacement; the same sampled query-track list is applied across the selected training seeds. Percentiles 2.5 and 97.5 over 10,000 draws define descriptive 95% intervals. The gallery of donors remains fixed. The separate 18-track motion-quality cohort uses common source-supported 23 s segments beginning at frame 264, three training seeds and three sampling seeds, yielding 162 outputs per configuration.

1. CONSTRUCT ONE GENERATED TRANSITION (STOP GRADIENT)



2. LEARN THE NEXT PLAN UNDER THE CONSTRUCTED CONTEXT

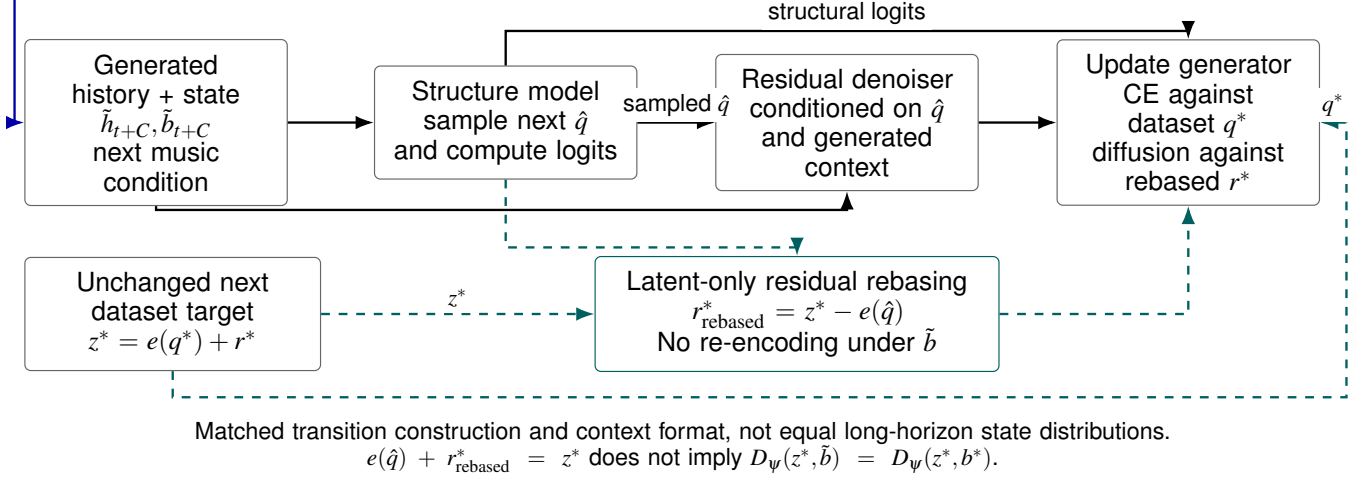


Fig. 6. **Commit Forcing transition construction and supervision.** A sampled, jointly committed structure–detail segment supplies the next history and reference boundary. The next dataset latent remains fixed; residual rebasing expresses it relative to the sampled next structure. This preserves the latent target, while the changed boundary can change its decoded motion. The two stages are constructed independently for each supplied training example.

Its scores and uncertainty are kept separate from the 60 s correspondence study.

A. Music–Motion Evaluator

a) Frozen representation and readout: The G1-native evaluator receives paired five-second windows: 150 audio feature frames at 30 Hz and 100 motion feature frames at 20 Hz. Motion features are extracted once from each complete trajectory, then normalized with frozen training-split statistics before windowing. The continuous 357-dimensional frontend retains full root rotation in $\text{SO}(3)$; it does not reset the orientation anchor separately for every scoring window. The two encoders produce deterministic posterior means for each window in FP32 inference. For evaluator instance k , denote these embeddings by $\mu_w^{x,k}$ and $\mu_w^{a,k}$. The adopted score is

$$s_w(x, a) = \frac{1}{3} \sum_{k=1}^3 \frac{\langle \mu_w^{x,k}, \mu_w^{a,k} \rangle}{\|\mu_w^{x,k}\|_2 \|\mu_w^{a,k}\|_2}. \quad (15)$$

Scores are averaged across the three independently trained evaluator instances before pairwise ranking; embeddings and final accuracies are not averaged instead. The checkpoints use standard content-group sampling and the final training update (14,500) for each seed, with no best-seed choice. Encoder weights and normalization remain fixed throughout generator scoring. An absolute score difference of at most

10^{-7} is treated as a tie and receives half credit. This specifies the correspondence statistic in Eq. (6) without conflating it with retrieval recall.

b) Evaluator validation data: Section IV-B reports the evaluator’s reference-motion results on 30 content groups and 3,661 windows. Each discrimination task uses 15 disjoint song pairs. These data are separate from the generated-motion gallery. The adopted score-level ensemble and all candidate comparisons are retained in the evidence record; calibration does not establish perceptual validity on executed robot trajectories.

c) Reproducible score aggregation: The reported generator configuration supplies nine training/sampling combinations, each with an $11 \times 11 \times 56$ music–motion score array. The diagonal contains intended pairs; each off-diagonal column supplies the ten alternative tracks for one fixed motion. All 99 per-motion scores, margins, and accuracies are reproduced from these arrays. No source or window is removed based on its score. The complete two-configuration archive and metric-specific reporting choices remain in the evidence record.

d) G1 distribution and local-motion features: Kinetic descriptors summarize motion dynamics, while geometric descriptors summarize body configurations and spatial relations in the G1 domain. FID compares feature means and covariances with those of retargeted references; diversity

measures pairwise distances in the same feature space. Extractor version, coordinate convention, normalization, crop length, and sample counts are part of each result. The earlier three-model table and current ForeDance results are not pooled: their numerical feature-scale equivalence has not been established. A multi-seed FID aggregation must retain how collections were formed; it is not a mean of single-song FIDs. For future execution scoring, generation and execution must use identical feature extraction. Exact feature lists and activity/contact thresholds require the corresponding implementation receipt.

e) Beat alignment: For motion-event times $\mathcal{B}_x$ and music-beat times $\mathcal{B}_a$, the motion-to-music BAS definition is

$$\text{BAS} = \frac{1}{|\mathcal{B}_x|} \sum_{b \in \mathcal{B}_x} \exp\left(-\frac{\min_{a \in \mathcal{B}_a} (b-a)^2}{2\sigma^2}\right). \quad (16)$$

The displayed BAS is the archived forward-kinematics motion-to-music score. At 30 Hz, mean keypoint speed is smoothed with a Gaussian of standard deviation five frames; strict local minima define motion beats. The matching kernel uses $\sigma = 3/30 = 0.10$ s. Music beats retain the archived audio-beat extraction and source offset. Empty music- or motion-beat sets receive zero. Scores are averaged over tracks and sampling repeats within each training seed, then across the three training seeds. The separately archived music-to-motion BAS uses the reverse matching direction and is not substituted into this column. Event counts, empty-event handling, smoothing, and any beat-matching tolerance must accompany reproducible scores. Beat precision/recall are auxiliary coverage diagnostics, not requirements that dance hit every audio beat. The layer analyzer’s impact proxy is a smoothed absolute derivative of the speed curve, not contact force. Zero-lag correlation with audio onset strength measures aligned dynamic response; best-lag correlation and its lag are diagnostics, not permission to retime primary music scores.

APPENDIX III

DETAILED CORRESPONDENCE STATISTICS

Table IV retains the component scores and margin intervals underlying Table II. Correct-minus-wrong means are positive for every generator training seed. These are the same 99 trajectories and the same fixed-gallery aggregation described in Appendix II. The companion scores and intervals use the same configuration as the displayed correspondence accuracy; they are not maximized independently. Intervals are descriptive conditional summaries of that configuration.

APPENDIX IV

REFERENCE DELIVERY AND SUPPORTING EXECUTION DIAGNOSTICS

a) Replay setup: Four generated trajectories from one FineDance track and one sampling seed are replayed in MuJoCo for approximately 60 s each.

b) Reference construction and delivery: A four-step latent commit contains eight 30 Hz frames. A continuous 50 Hz time grid yields 13 or 14 reference samples per commit without resetting the sampling phase at each packet. The controller receives committed references rather than motor torques. The four diagnostic replays use the same fixed SONIC controller at real-time playback, with no uncommitted-plan preview, clipping, startup ramp, or alignment hold. Initial state, delivered future-reference access, and packet timestamps are part of the protocol; full-trajectory delivery and incremental committed-motion delivery are not interchangeable controls. The trajectories begin at an artificial continuation point with empty motion history, rather than at full-song onset.

c) Observed layers: Let M_{gen} , M_{sent} , M_{target} , and M_{exec} denote generated, resampled, consumed, and measured motion. Target and measured joints use named fields in MuJoCo order, with matched coordinate conventions, offsets, and timestamps.

d) Conditions and measurements: The first three trajectories share cached forecast content but use different age traces: ideal age, the earlier measured trace, and a revised measured trace. The fourth uses live forecast content with the revised trace. These labels identify the generation conditions of the saved trajectories.

For active joint j , amplitude and velocity-variation retention are

$$R_j^{\text{amp}} = \frac{P_{95}(q_j^{\text{exec}}) - P_5(q_j^{\text{exec}})}{P_{95}(q_j^{\text{target}}) - P_5(q_j^{\text{target}})}, \quad (17)$$

$$R_j^{\text{dyn}} = \frac{\mathbb{E}[(\dot{q}_j^{\text{exec}})^2]}{\mathbb{E}[(\dot{q}_j^{\text{target}})^2]}. \quad (18)$$

Only joints with a target percentile range above 0.02 rad enter the active-joint summary, and ratios with near-zero denominators are excluded. The reported retention values are medians of joint-wise ratios. The dynamic ratio measures squared joint velocity; it is not physical energy or a whole-body energy ratio. Joint-position RMSE is evaluated after time alignment. Alignment lag is estimated within ± 0.5 s on the 50 Hz grid; the four recorded estimates are 40 ms, which is a case-specific alignment result rather than a general controller or system latency. Steady-state summaries exclude the first two seconds, so they do not establish startup behavior.

e) Interpretation: Amplitude retention ranges from 0.858 to 0.901, while squared-velocity retention ranges from 0.411 to 0.496. Position tracking thus coexists with dynamic attenuation. High-frequency variation can include articulation and jitter; these measurements alone do not determine the musical value of the attenuation.

TABLE IV

FULL SOUNDTRACK-CORRESPONDENCE STATISTICS. CORRECT AND WRONG DENOTE COSINE SCORES WITH INTENDED AND ALTERNATIVE GALLERY MUSIC. MARGIN IS CORRECT MINUS WRONG. PAIRWISE ACCURACY GIVES HALF CREDIT TO TIES; INTERVALS ARE DESCRIPTIVE 95% BOOTSTRAP INTERVALS WITH THE GALLERY FIXED.

Method	Correct	Wrong	Margin	Margin interval	Pairwise accuracy (%) [interval]
ForeDance	0.2057	0.0274	+0.1783	[0.0983, 0.2608]	75.01 [65.17, 84.13]

TABLE V

SUPPORTING FIXED-REFERENCE REPLAY MEASUREMENTS. DATA FROM THE EARLIER 34D GENERATOR, SEPARATE FROM THE CURRENT 38D COHORT. EACH ROW IS ONE MUJoCo REPLAY THROUGH FIXED SONIC, EXCLUDING ITS FIRST TWO SECONDS; MUSIC INFERENCE IS NOT RUN DURING REPLAY.

Reference-generation condition	Aligned RMSE (rad)	Median R^{amp}	Median R^{dyn}
Cached forecast, ideal age	0.1421	0.869	0.496
Cached forecast, earlier age trace	0.1281	0.858	0.428
Cached forecast, revised age trace	0.1394	0.877	0.412
Live forecast, revised age trace	0.1431	0.901	0.411

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